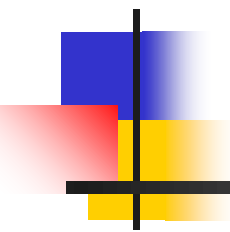


COMP 225-hybrid: Software Engineering Methodologies

Software Requirements Specification (SRS)

Name of project : VASDP: The Voice Activated Speaker for Disabled People



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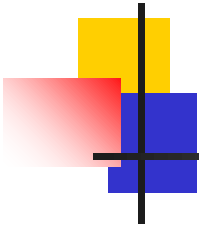
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Date : 08/06/2024

Section #: 005

Semester : 2



Introduction

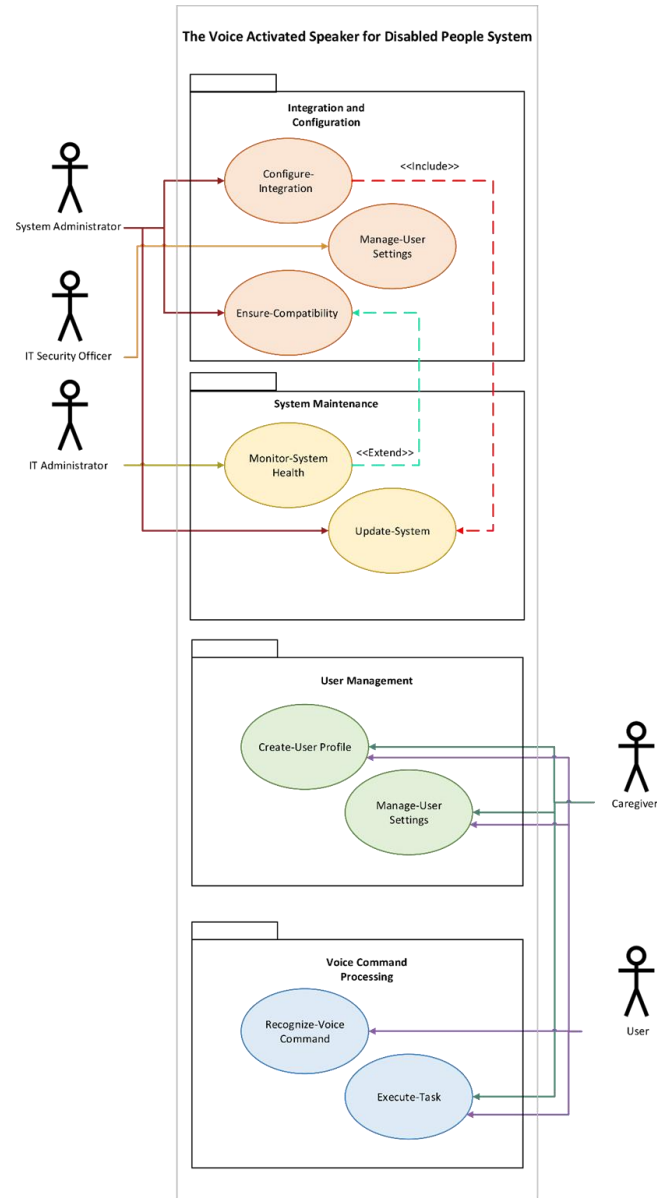
- Purpose

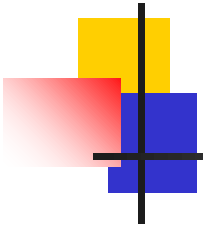
This new software product will address the problem of Disabled with the voice activated speaker and will be integrated into the infrastructure of Canada, this is version 1.0.0.

- Project scope

The Voice Activated Software Speaker for Disabled People (VASDP) aims to enhance accessibility for individuals with disabilities (excluding the deaf) by providing a voice-controlled interface for various tasks. This software will integrate seamlessly into Canada's existing technological infrastructure, offering a user-friendly solution to improve independence and quality of life for the disabled community.

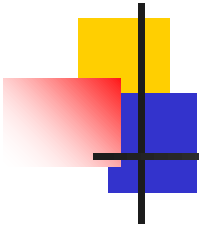
Packaged Use Case Diagram





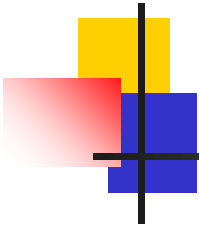
Use Cases Brief Descriptions only 6

Goal Use Case name	Actors	Use Case Brief Description
1) Create-User Profile	User, Caregiver	The actor creates a new user profile with personalized settings, including visual adjustments and voice command preferences.
2) Manage-User Settings	User, Caregiver	The actor updates user settings such as voice command preferences, color themes, and accessibility options.
3) Recognize-Voice Command	User	The system accurately recognizes and processes voice commands issued by the user, translating them into actions or responses.
4) Execute-Task	User, Caregiver	The system performs tasks such as setting reminders, controlling home devices, and accessing information based on recognized voice commands.
5) Configure-Integration	System Administrator	The actor configures the system to integrate with various external devices and services such as smart home systems and other assistive technologies.
6) Manage-Security	IT Security Officer	The actor ensures that all user data is encrypted and secure, configuring necessary security measures such as authentication and authorization protocols.



Use Cases Brief Description

Goal Use Cases	Use Case Brief Description
Ensure-Compatibility	The actor verifies and manages the system's compatibility with different hardware platforms, operating systems, and third-party services.
Monitor-System Health	The actor monitors system performance and uptime, ensuring the system is available and scales according to user demand.
Update-System	The actor updates the system with new features, security patches, and performance improvements to ensure continuous improvement in accessibility and functionality.



Use Case Full Description (present one)

Use Case Name: Recognize-Voice Command

Primary Actor: User (Disable people)

Stakeholders and Interests:

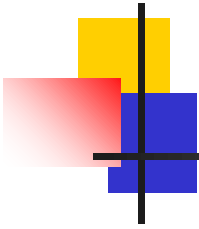
- **User** (Disable people) : Wants to interact with the system using voice commands for ease of use.
- **Caregiver:** Wants to ensure the system accurately recognizes commands to assist the user.
- **System Administrator:** Needs to ensure the system's voice recognition functions reliably and efficiently.

Preconditions:

- 1.The user is logged into the system.
- 2.The voice recognition system is initialized and running.
- 3.The microphone is functioning and properly configured.
- 4.The camera is properly set up and functioning if required.

Postconditions:

- 1.The system correctly interprets the user's voice command.
- 2.An appropriate response or action



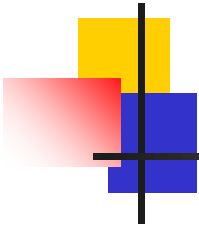
Use Case Full Description (present one)

Main Success Scenario:

- 1.The user issues a voice command.
- 2.The system captures the voice input via the microphone.
- 3.The system processes the captured voice input using the voice recognition engine.
- 4.The system interprets the voice command accurately.
- 5.The system provides feedback to the user, either confirming the command or executing the requested action

Extensions (Alternative Flows):

- **2a.** Microphone is not functioning:
 - 2a1. The system notifies the user of the microphone issue and prompts to check the device.
- **3a.** Camera is required for additional input but not functioning:
 - 3a1. The system notifies the user of the camera issue and prompts to check the device.
 - 3a2. The system offers to proceed with audio-only mode if camera input is optional.
- **4a.** Voice command is not recognized:
 - 4a1. The system requests the user to repeat the command.
 - 4a2. The system processes the repeated command.
 - 4a3. If the command is still not recognized, the system offers alternative input methods (e.g., text input).



Use Case Full Description (Continued)

Special Requirements:

- 1.The system must process voice commands with high accuracy, even with various accents and speech patterns.
- 2.The system should provide real-time feedback to ensure seamless interaction.

Technology and Data Variations List:

- 1.Voice recognition engine can be cloud-based (e.g., Google Cloud Speech-to-Text) or locally hosted.
- 2.Microphone can be built-in or external.
- 3.Camera can be built-in or external and used for optional visual input.

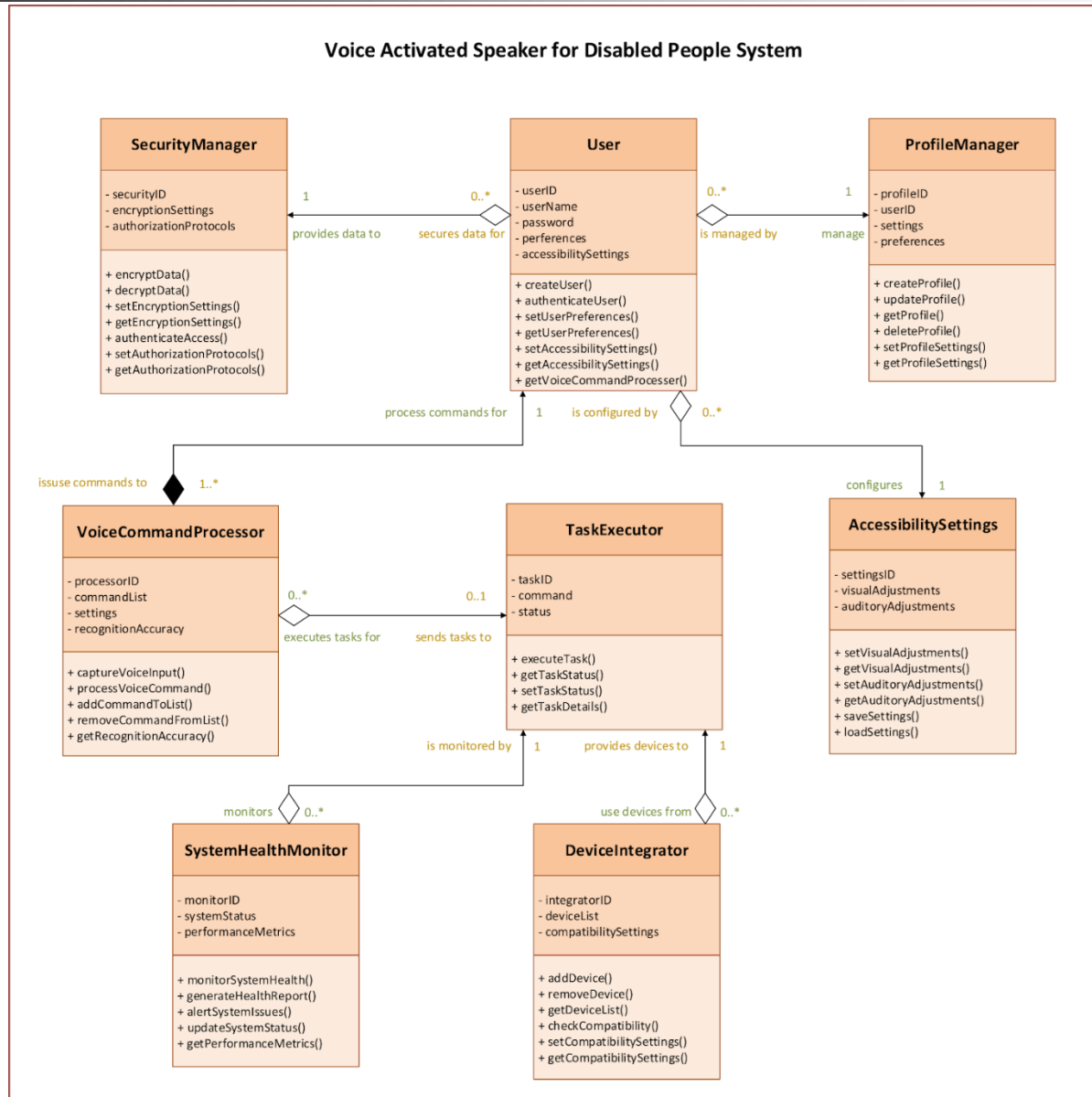
Frequency of Occurrence:

- Frequently, as it is the primary method for user interaction with the system.

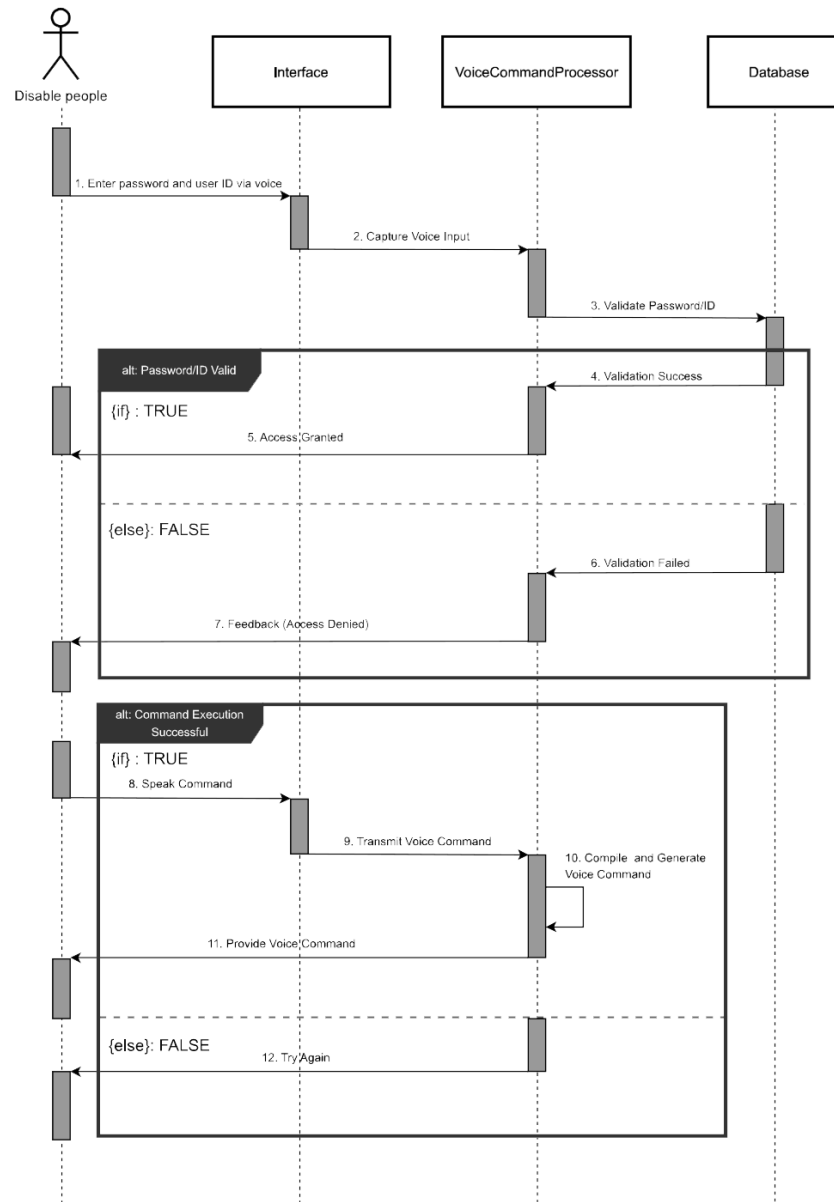
Open Issues:

- 1.Ensuring privacy and data security for voice and visual data.
- 2.Handling variations in speech, environmental noise, and lighting conditions for camera input.

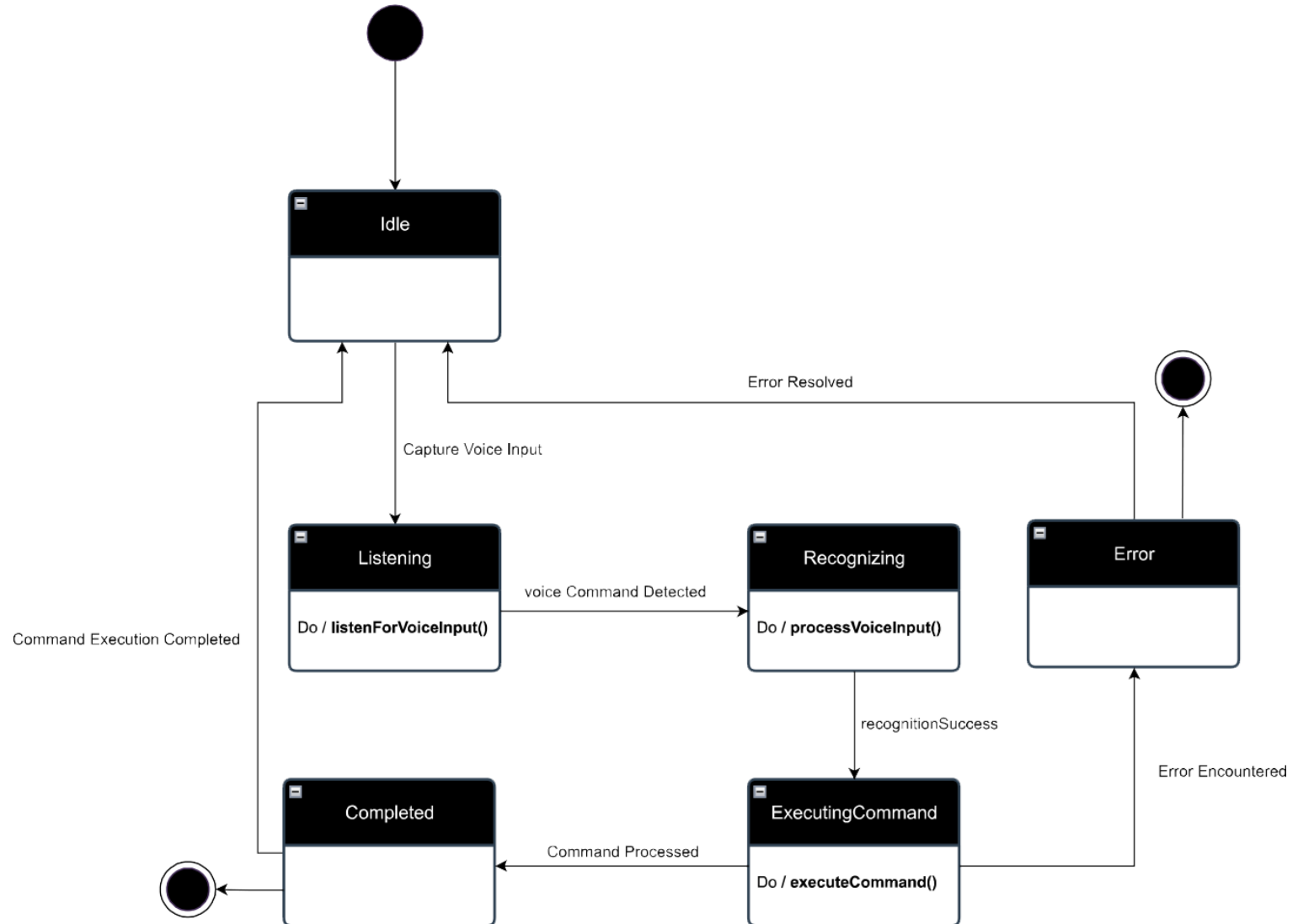
Domain Class Diagram with methods



Sequence diagram (present one)



State machine diagram



State machine diagram

